When Do Failures Reveal Spurious Structure?
A Study of Label-Free Latent-Risk Editing in Frozen Vision-Language Models

Anonymous submission

Abstract

Errors of frozen vision-language models (VLMs) can concentrate in groups that are unobserved during adaptation. We ask whether task failures alone expose an editable latent-risk subspace without group or sensitive-attribute labels. LatentRisk-VLM converts task-labeled calibration margins into risk labels, removes class-level semantics, and estimates class-conditional risk directions by residual mean contrast or iterative linear probes. The subspace is defined operationally by failure predictiveness, weak semantic coupling, and suppressibility; it is not a discovered sensitive attribute. Experiments with frozen CLIP ViT-L/14 show a conditional rather than universal benefit. On Waterbirds, task-label-only rank-4 editing reaches 64.1 WGA in a three-seed rerun, while DFR-Frozen reaches 90.2 WGA with full attribute labels. On CelebA, rank-1 editing lowers WGA from 71.5 to 54.4 despite raising average accuracy from 78.2 to 91.7, while rank-4 worst-class selection reaches 72.8 WGA. Attribute-budget and concept-erasure experiments sharpen the interpretation: LatentRisk-VLM occupies the zero-attribute-label point, whereas explicit attribute labels remain highly beneficial when available. INLP raises Waterbirds WGA only to 43.1 and LEACE leaves WGA unchanged in our global feature-space erasure, and the measured risk–attribute subspace overlap is low. Additional low-cost extensions with frozen SigLIP and a MetaShift-style cat/dog contextual shift preserve the same cautionary conclusion: group-free task-accuracy selection can fail badly, while rank-4 worst-class selection can recover WGA only when the candidate pool and diagnostic evidence are favorable. Failure-derived directions are therefore useful weak supervision when failure and group shift align, but neither risk separability nor task accuracy guarantees group robustness.

Introduction

Frozen vision-language models (VLMs) provide transferable image–text representations for classification and retrieval (Radford et al. 2021). Their zero-shot interface, however, does not remove biases inherited from web-scale pretraining. Errors can concentrate on demographic groups (Agarwal et al. 2021; Hall et al. 2023), but also on backgrounds, acquisition conditions, pose, or other spurious factors. Standard average accuracy can conceal these failures.

Most post-hoc VLM debiasing methods assume that the relevant attribute is known. They encode named concepts in prompts (Chuang et al. 2023), use attribute-labeled reference images (Gerych et al. 2024), or estimate attribute-predictive subspaces (Zhao et al. 2026). This is appropriate when the attribute is available and is the actual source of harm. In many deployments, however, group labels are unavailable, restricted, incomplete, or simply miss the factor driving errors. What remains observable in deployment is the model's pattern of errors and low-confidence predictions.

We therefore ask: *can low classification margins—where the correct-class score is close to that of the strongest competing class—and prediction errors reveal a representation direction whose removal reduces hidden-group failure?* This question differs from discovering a demographic attribute. A failure-predictive direction may encode background, image quality, an intersection of factors, or useful class evidence. Consequently, editing it may improve worst-group reliability, do nothing, or make performance less balanced. A credible method must expose this distinction rather than equating any failure direction with a fairness direction.

We call a low-dimensional subspace a *latent risk subspace* when it is predictive of within-class calibration failures, weakly coupled to class semantics, and suppressible without destroying the task score. This is an operational representation of error concentration, not a recovered gender, race, age, or other sensitive attribute. If it overlaps the factor defining a worst-performing group, suppressing it can indirectly improve worst-group performance; if it does not, editing can be harmful.

We introduce LatentRisk-VLM, a post-hoc framework for frozen VLM embeddings. Given a task-labeled calibration set, we derive risk labels from zero-shot margins and remove class prototypes and text-aligned evidence. We estimate a class-conditional risk direction either by contrasting high- and low-risk residual means or by fitting iterative linear probes inspired by INLP (Ravfogel et al. 2020). Test labels are not used: each candidate class supplies its own prototype and subspace, producing candidate-wise corrected scores. All selection in the primary protocol uses task labels only; sensitive attributes are read after prediction solely for group metrics.

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The completed experiments give a deliberately mixed answer. On Waterbirds, rank-1 editing improves WGA by 26.9 points without place labels, although it loses 5.3 points of average accuracy. Rank-4 editing with group-free worst-class validation selection reaches 63.1 WGA on Waterbirds and 72.8 WGA on CelebA Blond Hair. However, other group-free selectors over the same candidate pool perform much worse, and attribute-labeled validation still raises rank-4 CelebA WGA to 82.2. A post-hoc diagnostic explains the rank-1 contrast: Waterbirds risk scores are enriched for minority-place groups, while CelebA risk directions have little alignment with the Male attribute direction. A frozen-SigLIP extension preserves the cautionary part of the story: task-accuracy selection fails on both datasets, while rank-4 worst-class selection improves WGA relative to a weak frozen SigLIP baseline but often pays a large average-accuracy cost. On a MetaShift-style cat/dog indoor/outdoor shift, a preregistered diagnostic predicts uncertainty and the final held-out gain is positive but small (+5.6 WGA points). Discovering a risk direction and selecting a group-robust operating point are therefore separate problems; CnC-Frozen with attribute-labeled validation reaches 85.3. A FairFace/Flickr30K retrieval audit likewise finds a modest skew reduction accompanied by a 4.0-point Recall@10 loss.

Our contributions are fivefold. First, we formulate latent-risk editing in frozen VLMs using task labels and failure signals, but no group or sensitive-attribute annotations, to estimate class-conditional risk directions. At test time, because the true class is unknown, the method edits and scores each candidate class separately rather than selecting an editor using the ground-truth label. Second, we develop an attribute-label-free, post-hoc low-rank editing procedure that leaves both VLM encoders unchanged and runs on cached features. Third, we add an attribute-label budget protocol that exposes the zero-annotation operating point and separates training supervision from model-selection supervision. Fourth, we compare against supervised concept-erasure baselines such as INLP and LEACE to test when failure-derived directions substitute for a named attribute direction and when they do not. Fifth, under a shared frozen-backbone setting, we compare the additional supervision, optimization, and adaptation cost required after feature extraction. The contrasting results are therefore central to our conclusion: failures provide useful weak supervision when they align with hidden groups, but they do not consistently identify those groups.

Related Work

VLM bias measurement and post-hoc editing. Audits have documented demographic and stereotype-related disparities in VLM classification and retrieval (Agarwal et al. 2021; Hall et al. 2023; Zhou, Lai, and Jiang 2022; Hamidieh et al. 2024). Prompt and projection methods remove directions specified by biased prompt pairs (Berg et al. 2022; Chuang et al. 2023); BEND-VLM uses attribute-labeled references for query-specific correction (Gerych et al. 2024); SANER learns a societal-attribute neutralizer from a predefined lexicon without per-image attribute labels (Hirota et al. 2025); and recent work treats bias as an attribute-predictive subspace (Jung, Jang, and Wang 2024; Zhao et al. 2026). These methods start from an attribute or concept. We instead start from within-class failure and test post hoc whether the resulting editor improves group metrics.

Robustness without training-group labels. Group DRO directly optimizes worst-group risk when group annotations are available (Sagawa et al. 2020). JTT uses early errors to up-weight hard examples (Liu et al. 2021), EIIL infers environments from loss (Creager, Jacobsen, and Zemel 2021), and last-layer methods improve robustness with limited group supervision (Kirichenko, Izmailov, and Wilson 2023; Qiu et al. 2023). Correct-N-Contrast forms contrastive sets from error-derived groups (Zhang et al. 2022). Several methods avoid group labels during fitting but still use group-labeled validation data for hyperparameter or checkpoint selection; we distinguish this regime from both fully group-free selection and attribute-supervised adaptation in Table 1. Our setting is more restrictive in model access: image and text encoders remain frozen, and intervention occurs in their shared embedding space. It is also diagnostically different: we do not assume that errors recover the true groups, and our CelebA result demonstrates the cost of that assumption.

Representation subspaces. INLP iteratively removes linearly decodable information (Ravfogel et al. 2020), and LEACE gives a closed-form linear concept-erasure solution (Belrose et al. 2023). Recent VLM work also argues that social bias is distributed across subspaces rather than individual coordinates (Zhao et al. 2026). We evaluate both a transparent mean-contrast direction and iterative risk probes. On Waterbirds, mean contrast outperforms the rank-4 construction; on CelebA, even a highly risk-separable rank-1 probe harms WGA. Our focus is therefore not whether risk is decodable, but whether suppressing it preserves task semantics and improves held-out groups.

Difference from supervised concept erasure. Concept erasure begins with a named concept and learns how to remove it from a representation. INLP and LEACE therefore answer: how can a known concept be made linearly inaccessible? LatentRisk-VLM begins instead with model failures and asks which latent directions are repeatedly activated by those failures. The resulting direction may overlap with a spurious attribute, but no attribute identity is assumed or required. We do not claim to recover the true sensitive attribute; a risk direction is only a failure-related feature pattern. If it aligns with the annotated attribute, failure supervision may approximate attribute-supervised erasure. If it only partially aligns, it may remove the failure-relevant component while retaining other task semantics. If alignment is low, it may capture factors beyond the named spurious attribute.

LatentRisk-VLM

Problem Setup

Let f_I and f_T be frozen image and text encoders. For image x_i and class prompt p_c, the unit-normalized embeddings are $z_i = f_I(x_i) \in \mathbb{R}^d$ and $t_c = f_T(p_c) \in \mathbb{R}^d$. Frozen zero-shot prediction is

$$\hat{y}_i^{\text{zs}} = \arg\max_{c \in \mathcal{Y}} z_i^\top t_c. \tag{1}$$

We observe a calibration set $D_{\mathrm{cal}} = \{(x_i, y_i)\}_{i=1}^n$ with task labels but no group attributes. A group attribute a_i (place in Waterbirds or gender in CelebA) may be revealed only after model selection to compute group accuracy. The primary protocol forbids a_i during direction discovery, intervention, hyperparameter selection, and test-time scoring.

The calibration margin and risk are

$$m_i = z_i^\top t_{y_i} - \max_{c \neq y_i} z_i^\top t_c, \tag{2}$$

$$\rho_i = \sigma(-m_i/\tau), \qquad h_i = \mathbb{I}[\rho_i \geq \eta]. \tag{3}$$

A negative margin is a zero-shot error, so with the measured setting $\eta = 0.5$, h_i separates errors from correct predictions, where $h_i = 1$ indicates an incorrect zero-shot prediction and $h_i = 0$ indicates a correct zero-shot prediction. We seek a class-conditional, low-rank representation U_c with three properties: it predicts h_i after class semantics are removed, has weak coupling to the class score, and can be suppressed without erasing the score needed for class c. These properties define a latent risk subspace; they do not imply that U_c encodes a named sensitive attribute.

At test time y is unknown. "Class-conditional" therefore cannot mean selecting U_y with the ground-truth label. For every candidate c, the method constructs a corrected embedding $\hat{z}^{(c)}$ and score $s_c(x)$, then predicts $\arg\max_c s_c(x)$. This candidate-wise rule is part of the problem definition and prevents test-label leakage.

Figure 1 shows the complete pipeline. The backbone is used only to encode images and text; all subsequent operations occur in its normalized embedding space.

Removing Class-Level Semantics

A risk estimator applied directly to z_i may rely on ordinary class-discriminative evidence rather than features genuinely associated with classification error. For each class c, we therefore compute a class prototype, defined as the mean embedding of its calibration samples:

$$\mu_c = \frac{1}{|D_c|} \sum_{i: y_i = c} z_i, \tag{4}$$

and form a residual orthogonal to the class text embedding:

$$r_i = z_i - \mu_{y_i}, \qquad \tilde{z}_i = r_i - t_{y_i} t_{y_i}^\top r_i. \tag{5}$$

Subtracting the class prototype restricts risk-direction discovery to within-class variation, while the second operation reduces the direct coupling between the representation and the corresponding class-prompt direction. However, these operations do not guarantee that all class information has been completely removed.

Unified design objective. We aim to identify a risk direction, or risk subspace, U_c that captures error-related information while containing as little class-semantic information as possible, such that suppressing it does not impair the classification capability. A compact specification is

$$\begin{aligned} \min_{U_c,\, \hat{z}^{(c)}} \quad & \mathcal{L}_{\mathrm{risk}}(U_c; \tilde{Z}_c, h_c) + \alpha \|t_c^\top U_c\|_2^2 \\ & + \lambda \|U_c U_c^\top (z - \mu_c)\|_2^2 - \hat{z}^{(c)\top} t_c. \end{aligned} \tag{6}$$

Here, $U_c^\top U_c = I$,$\mathrm{rank}(U_c) \leq r$, $\hat{z}^{(c)} = \mathrm{Edit}(z; U_c, \ell_c)$.

The first term encourages the learned U_c to predict calibration risk. The second discourages coupling between the risk subspace and the class semantics. A large value of $\|t_c^\top U_c\|_2^2$ indicates that the learned risk directions are strongly aligned with the class text embedding, suggesting that the model may be capturing class-discriminative information rather than risk-related information. The third term penalizes the candidate's activation in the risk subspace. Specifically, $U_c U_c^\top (z - \mu_c)$ projects the class-centered feature onto the risk subspace. A large norm indicates that the candidate contains a strong risk component; therefore, the objective encourages this activation to be reduced. The fourth term preserves task semantics by encouraging the edited image embedding $\hat{z}^{(c)}$ to remain highly similar to the class text embedding t_c.

Regarding the constraints, $U_c^\top U_c = I$ requires the columns of U_c to be orthonormal. In other words, for $U_c = [u_1, u_2, \ldots, u_r]$, we have $u_i^\top u_j = 0$ for $i \neq j$ and $u_i^\top u_i = 1$. The constraint $\mathrm{rank}(U_c) \leq r$ restricts the search to a low-dimensional risk subspace. Finally, $\hat{z}^{(c)} = \mathrm{Edit}(z; U_c, \ell_c)$, where Edit denotes the normalized candidate-wise projection and reinjection defined in Equations 8–10: the risk component is first removed by projection, after which neutral information is reinjected.

This formulation is a design objective rather than an additional jointly optimized loss. In practice, class residualization reduces semantic coupling, mean contrast or Risk-INLP estimates the risk subspace, projection implements suppressibility, and neutral reinjection together with the semantic score preserves utility. The scoring grid determines the operating point, and α is not tuned separately.

Estimating Risk Directions

Let $H_c = \{i : y_i = c, h_i = 1\}$ denote the set of high-risk or misclassified samples from class c, and let $L_c = \{i : y_i = c, h_i = 0\}$ denote the corresponding low-risk or correctly classified samples. We estimate a class-specific risk direction by contrasting the mean residualized embeddings of these two groups:

$$u_c = \frac{\bar{z}_{H_c} - \bar{z}_{L_c}}{\|\bar{z}_{H_c} - \bar{z}_{L_c}\|_2}, \quad \bar{z}_S = \frac{1}{|S|} \sum_{i \in S} \tilde{z}_i. \tag{7}$$

Here, $\bar{z}_{H_c}$ and $\bar{z}_{L_c}$ are the centroids of the high-risk and low-risk samples, respectively, in the residualized feature space. Their difference therefore points from the average representation of successful predictions toward that of failures. Normalizing this difference removes its scale and yields a unit vector that captures only the dominant direction of risk-related variation within class c.

Thus, $U_c = u_c \in \mathbb{R}^{d \times 1}$ defines a one-dimensional, class-conditional risk subspace. For a residualized embedding $\tilde{z}_i$, the projection $u_c^\top \tilde{z}_i$ measures its activation along this direction, with larger values indicating greater alignment with the feature pattern associated with failures. The mean-contrast estimator provides the simplest linear description of how failures differ from successes after class residualization. Since it is constructed directly from the separation between high-risk

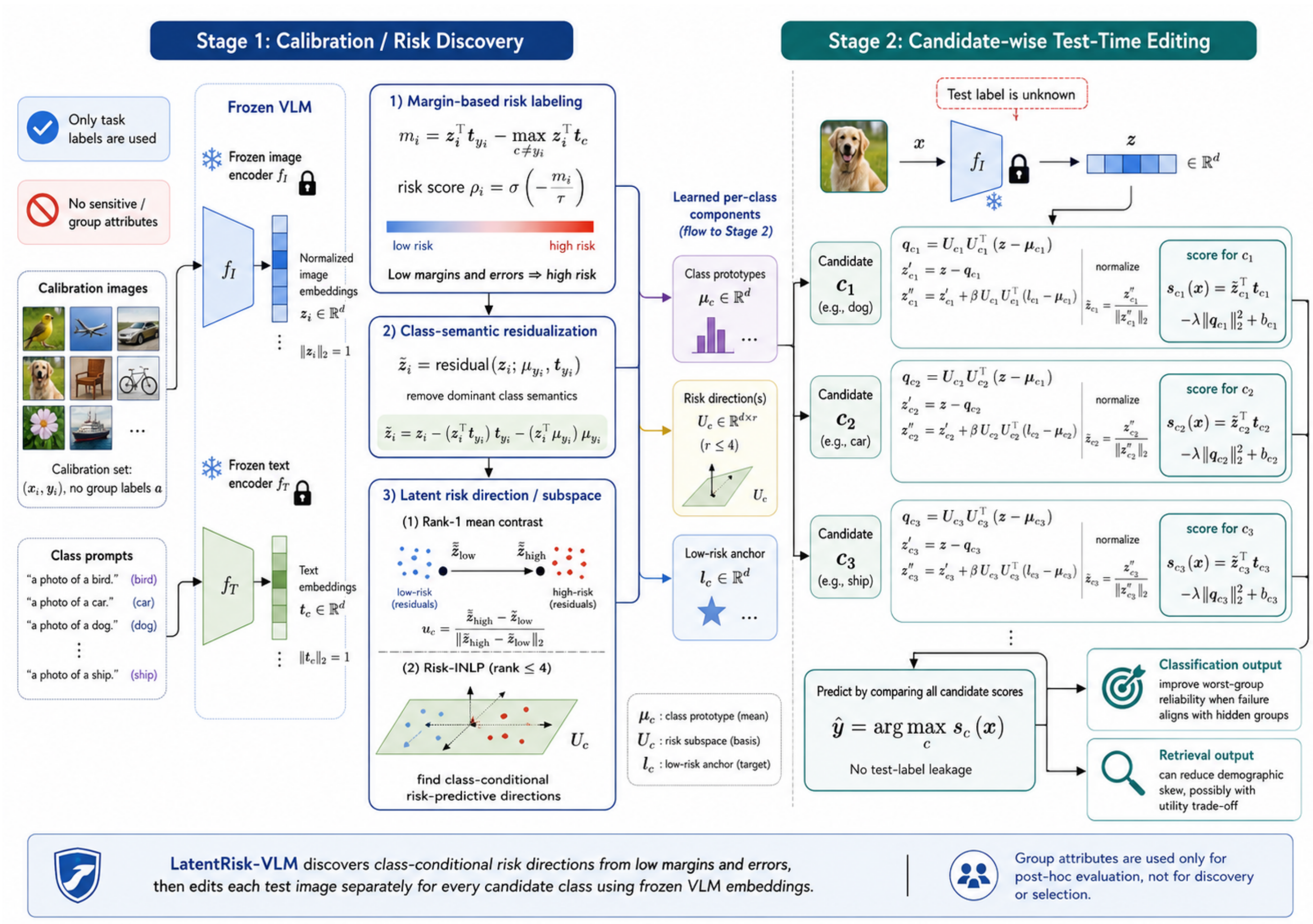

Figure 1: LatentRisk-VLM. Risk directions are estimated from high- and low-risk residuals using mean contrast or iterative probes. At test time every candidate class is scored with its own correction; no test label selects the edit.

and low-risk samples, it is risk-predictive by design. However, because no demographic or sensitive-attribute annotations are used, the resulting direction should be interpreted only as a latent risk direction rather than as a named demographic direction.

Probe-based directions. Alternatively, we fit a linear logistic probe to h_i within each class. Its normalized weight becomes $u_{c,1}$; the residuals are projected into its null space, and the process repeats up to rank four or until probe accuracy falls below 0.55. QR orthogonalization yields $U_c = [u_{c,1}, \ldots, u_{c,r}]$. This Risk-INLP estimator follows the iterative removal principle of Ravfogel et al. (2020). Rank one tests a single discriminative direction, while rank four tests whether residual risk remains distributed after successive projections.

Candidate-Wise Representation Editing

For a test embedding z and candidate class c, define its residual risk component $q_c = U_c U_c^\top (z - \mu_c)$. Full projection gives

$$z'^{(c)} = z - q_c. \tag{8}$$

Projection may move an embedding away from regions represented by low-risk calibration examples. Let ℓ_c be the mean original embedding over L_c. We reinsert half of its risk-subspace component,

$$z''^{(c)} = z'^{(c)} + \beta U_c U_c^\top (\ell_c - \mu_c), \qquad \beta = 0.5, \tag{9}$$

and normalize $\hat{z}^{(c)} = z''^{(c)} / \|z''^{(c)}\|_2$. The candidate score is

$$s_c(x) = \hat{z}^{(c)\top} t_c - \lambda \|q_c\|_2^2 + b_c. \tag{10}$$

The first term retains CLIP semantic matching, the second penalizes candidate-specific risk activation, and b_c is a class intercept. Prediction compares Equation 10 over all c; it never applies a correction chosen by the ground-truth test class.

Task-Label-Only Selection

We choose λ and b_c on calibration data after direction discovery. Each candidate stores only sanitized validation records: sample identifier, task label, frozen-baseline prediction and scores, baseline true-class margin, candidate prediction and scores, and candidate loss. Group or attribute fields are excluded from the records passed to group-free selectors. We evaluate five task-label-only selectors: overall task accuracy; worst-class accuracy, $\min_c \mathrm{Acc}(y = c)$; high-risk accuracy on the fixed lowest-margin fraction under the frozen baseline; net correction, which rewards baseline errors fixed by the candidate and penalizes baseline-correct samples broken by the candidate; and a min–max normalized combination of high-risk accuracy, worst-class accuracy, and net correction. The default high-risk fraction is 0.20, net-correction uses penalty weight 1.0, and the combined selector uses weights 1.0, 0.5, and 0.5.

All five selectors depend on y but not a. Ties are broken deterministically by validation task accuracy, candidate loss, rank, intervention size, and candidate identifier. For diagnosis, we also rerun the same search using validation WGA over (y, a) and label the resulting row *attribute oracle*. Comparing these selectors isolates a practical question: even if error-derived directions contain group-relevant information, can an attribute-free criterion choose a useful operating point?

Complexity. Using cached embeddings of dimension d, rank-1 discovery requires one pass over n calibration samples, $O(nd)$ time and $O(nd)$ storage. Candidate-wise inference costs $O(|\mathcal{Y}|d)$ per image beyond the encoder call. The rank-r extension increases projection cost to $O(|\mathcal{Y}|dr)$ and fits at most r linear probes per class. No variant adds a VLM forward pass or modifies backbone parameters.

Retrieval Adaptation

For Flickr30K calibration captions, risk is the negative margin between the paired image and the hardest nonmatching image. Paired image embeddings are centered by their global prototype and residualized against their caption embeddings. Splitting at the median margin yields one high–low risk direction. The same projection and reinjection edit every gallery image, while λ is selected by validation text-to-image Recall@10. The fitted editor is then evaluated unchanged on Flickr30K test retrieval and on FairFace demographic skew; FairFace gender labels never enter fitting or ranking.

Experimental Protocol

Backbone and reporting. The main experiments use 768-dimensional, normalized embeddings from the Transformers CLIP ViT-L/14 implementation. The low-cost extension additionally evaluates frozen SigLIP Base Patch16/224 (Zhai et al. 2023). In every case, both image and text encoders are frozen; the method operates only on cached shared-space embeddings. We report one deterministic run for each primary configuration. Results are measured adaptations to the stated frozen backbone, not numbers copied from the baseline papers.

Frozen-feature cost comparison. All methods in the controlled cost comparison share the same frozen CLIP backbone, the same cached image features, and the same calibration/evaluation split. Feature extraction time is reported separately from post-feature adaptation time. Under this shared frozen-backbone setting, we compare the additional supervision, optimization, and adaptation cost required after feature extraction. For controlled comparison, we adapt CnC and DFR to cached frozen CLIP features. These variants should not be interpreted as exact reproductions of the original end-to-end training procedures; they isolate the adaptation strategy under the same frozen representation. We therefore write CnC-Frozen and DFR-Frozen when referring to these feature-space variants, and "-Frozen" denotes feature-space adaptation using fixed CLIP embeddings.

Classification datasets. Waterbirds (Sagawa et al. 2020) uses the official validation split (1,199 images) for risk discovery and parameter selection and the test split (5,794 images) for evaluation. Fixed prompts are "a photo of a landbird" and "a photo of a waterbird." Groups cross bird type with place. CelebA (Liu et al. 2015) uses Blond Hair as the task and Male as the evaluation attribute. Its validation split (19,867 images) calibrates LatentRisk-VLM; its training split is available to fitted baselines; and its test split (19,962 images) is evaluated once. Prompts describe a person with or without blond hair. For the third classification setting, we build a deterministic MetaShift-style cat/dog contextual shift from GQA scene graphs (Hudson and Manning 2019) and Visual Genome images (Krishna et al. 2017), following the contextual-shift motivation of MetaShift (Liang and Zou 2022). The group attribute is indoor versus outdoor context. The split contains 1,500 train, 300 validation/calibration, and 144 test images; validation and test are balanced over the four class-context cells.

Retrieval datasets. FairFace (Karkkainen and Joo 2021) supplies a 10,954-image validation gallery and the 25 gender-stereotype queries released with BEND-VLM. Its 86,744-image training split is used only as the gender-labeled reference bank for BEND-VLM. We report text-to-image retrieval on the Flickr30K (Young et al. 2014) Karpathy test split (Karpathy and Fei-Fei 2015): 5,000 captions and 1,000 images. The 1,014-image validation split fits the LatentRisk retrieval editor. SANER is trained for five epochs on 124,026 person-related caption pairs from 27,576 disjoint Flickr30K training images.

Supervision accounting. We separate four classification settings according to where group attributes a enter the pipeline. *No example labels* includes frozen CLIP and prompt ensembling. *Task labels only (group-free adaptation and selection)* includes ERM, JTT-style calibration, and the primary LatentRisk-VLM variants; these methods use y but not a for either fitting/adaptation or hyperparameter selection. *Group-free adaptation, group-aware selection* includes CFR, CnC-Frozen, GIC, AFR, and our attribute-oracle selector: their reported fitting or editor construction does not use true group labels, but the operating point or checkpoint is selected with validation WGA computed from (y, a). *Group-aware adaptation and selection* includes SPD and BEND-VLM, which use attribute information during direction/reference construction and also retain attribute-aware validation. Group labels read only after prediction to compute test WGA do not affect this categorization. This accounting distinguishes learning an intervention without a from selecting among interventions with a.

Baselines. Classification evaluates frozen CLIP, prompt ensembling, ERM, JTT (Liu et al. 2021), AFR (Qiu et al. 2023), CnC-Frozen (Zhang et al. 2022), BEND-VLM (Gerych et al. 2024), SPD (Zhao et al. 2026), CFR (You et al. 2024), and a GIC adaptation. The reviewer-risk extension additionally includes DFR-Frozen (Kirichenko, Izmailov, and Wilson 2023) when a feature-space classifier refit is available, and INLP/LEACE concept erasure on cached embeddings. Retrieval evaluates frozen CLIP, projection prompts (Chuang et al. 2023), BEND-VLM, SANER (Hirota et al. 2025), and LatentRisk-VLM. Where a reference method originally trains a different backbone, we retain its central objective and selection rule but operate on CLIP ViT-L/14 features or lightweight adapters. This creates a controlled frozen-backbone comparison, not an exact reproduction of each paper's original setting.

Hyperparameters. Classification uses $\tau = 1$, $\eta = 0.5$, full projection, and $\beta = 0.5$. The measured CLIP rank-1 configuration uses mean contrast on Waterbirds and one logistic probe on CelebA; rank four uses iterative probes with $C = 1$ and a 0.55 training-accuracy stopping threshold. Waterbirds searches 101 values of $\lambda \in [0, 5]$ and 61 intercepts in $[-0.03, 0.03]$; CelebA searches 51 values in $[0.05, 0.30]$ over the same intercept grid. Rank-4 classification reports the full group-free selector sweep over this fixed candidate pool; the main group-free rank-4 row uses worst-class validation selection. The SigLIP extension keeps the same frozen-feature protocol, runs only frozen, rank-1, rank-4 worst-class, and rank-4 validation-WGA oracle settings, and reports WGA gains relative to frozen SigLIP. The MetaShift-style setting uses SigLIP, rank four, iterative probes, $\lambda \in [0.05, 0.30]$, the same intercept grid, and group-free worst-class validation selection. Retrieval searches 61 values of $\lambda \in [0, 0.3]$ and selects 0.1. The Waterbirds ablation changes one component at a time under group-free, task-label-only macro-accuracy selection.

Diagnostic prediction protocol. For the new MetaShift-style shift, we generate the diagnostic prediction before held-out test editing. A calibration-only run writes risk scores, residuals, and risk subspaces, then computes split-half stability, risk concentration, and risk-probe AUC without test labels. Post-hoc validation alignment and minority enrichment use validation group labels only for diagnosis. A fixed YAML rule maps these diagnostics to *likely success*, *uncertain*, or *likely failure*; the resulting `prediction.json` explicitly records that test performance was not used. Only after this file is written do we evaluate frozen SigLIP and final rank-4 editing on the test split.

Metrics. Classification reports average accuracy and WGA over the four (y, a) groups. We also discuss their difference as the average–worst gap. Retrieval reports text-to-image Recall@10 and mean gender MaxSkew@100 over the 25 FairFace queries. MaxSkew compares retrieved gender proportions with a uniform target; lower is better. For the first two settings, group attributes are accessed only after prediction to compute audit metrics. The remaining settings use a during selection or adaptation as stated explicitly in Table 1.

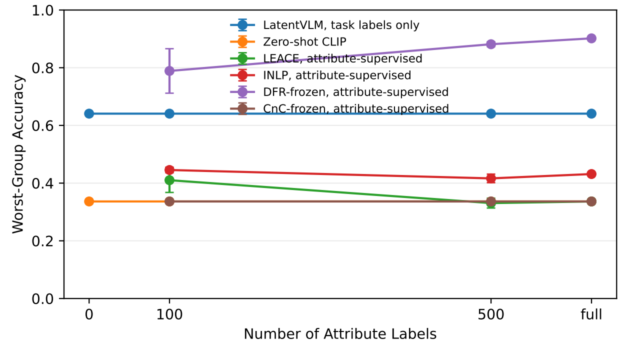

Figure 2: Waterbirds WGA as a function of available attribute labels. Attribute-supervised methods are not plotted at zero labels.

Measured Results and Analysis

Classification

Table 1 reports the measured classification results while separating group-label use in adaptation from group-label use in model selection. On Waterbirds, rank-1 LatentRisk-VLM reaches 60.5 WGA, improving frozen CLIP by 26.9 points while reducing average accuracy by 5.3 points. Rank-4 editing with group-free worst-class validation selection reaches 63.1 WGA, within 1.0 point of its validation-WGA oracle at 64.1. When validation group labels are allowed only for selection, AFR reaches 61.8 and the rank-4 oracle reaches 64.1. These oracle-selected rows should not be interpreted as group-label-free end-to-end pipelines, because their operating points are chosen using validation WGA. The Waterbirds result therefore supports failure-derived editing when the nuisance is unnamed, while exposing an explicit robustness–utility tradeoff.

CelebA remains selection-sensitive. Rank-1 editing raises average accuracy from 78.2 to 91.7 but lowers WGA from 71.5 to 54.4. Rank-4 worst-class selection improves to 72.8 WGA, while attribute-oracle selection reaches 82.2 WGA at 85.7 average accuracy. Because the editor is learned without Male labels in both rank-4 rows, this comparison isolates the contribution of group-aware selection rather than group-aware representation discovery. The 82.2 result is therefore an oracle upper bound, not an attribute-free robustness result, and it remains below the 85.3 WGA of CnC-Frozen under group-aware selection.

Attribute-Label Budget

Figure 2 and Table 2 show the intended operating-regime distinction. At zero attribute labels, attribute-supervised methods are not applicable, while LatentRisk-VLM-UL reaches 64.1 WGA with task labels and failure signals only. Once attribute labels are available, DFR-Frozen is substantially stronger: it reaches 78.9 WGA with 100 sampled attribute labels, 88.1 with 500, and 90.2 with the full validation attribute set. The budget-aware CnC-Frozen adapter does not improve over frozen CLIP in this rerun, staying at 33.6 WGA across nonzero budgets, so we do not copy its stronger single

Table 1: Classification with frozen CLIP ViT-L/14 encoders, grouped by where labels enter the pipeline. WGA, Avg., and Gap are worst-group accuracy, average accuracy, and their difference (percentage points). Using group labels only for validation-WGA selection is reported separately from using them during adaptation or reference construction.

		Waterbirds			CelebA		
Label setting	Method	WGA	Avg.	Gap	WGA	Avg.	Gap
No example labels	Frozen CLIP	33.6	83.7	50.1	71.5	78.2	6.7
	Prompt ensemble	41.1	83.8	42.7	65.0	77.2	12.1
Task labels only group-free adaptation and selection	ERM	29.4	**86.6**	57.2	16.1	**94.3**	78.2
	JTT-style calibration	36.1	71.5	35.3	31.1	84.5	53.4
	LatentRisk-VLM (rank 1)	60.5	78.4	17.9	54.4	91.7	37.3
	LatentRisk-VLM (rank 4, worst-class sel.)	**63.1**	79.5	**16.4**	72.8	87.9	15.1
Group-free adaptation group-aware selection	CFR	47.7	74.4	26.7	9.7	39.0	29.3
	CnC-Frozen	32.9	83.8	50.9	**85.3**	87.5	**2.3**
	GIC-Cy + Subsample	58.7	78.8	20.1	28.3	90.6	62.3
	AFR	61.8	80.2	18.3	70.6	88.2	17.7
	LatentRisk-VLM (attr.-oracle selection)	64.1	79.9	15.8	82.2	85.7	3.5
Group-aware adaptation and selection	SPD (oracle attr.)	22.7	77.2	54.4	67.7	77.8	10.2
	BEND-VLM	31.0	81.2	50.2	58.6	73.5	14.9

Table 1 configuration into the budget curve. This is not a failure of the comparison protocol; it is the point of the budget experiment. Explicit attribute labels remain highly valuable when available, and LatentRisk-VLM trades that supervised robustness for a zero-attribute-label operating point.

Table 3 shows that supervised concept erasure and failure-derived risk editing are not the same operation in this setting. LEACE leaves Waterbirds WGA unchanged at 33.6 in the global feature-space erasure, while INLP reduces attribute probe accuracy from 94.9 to 70.6 and improves WGA to 43.1. The oracle mean-difference direction, which directly uses the annotated background attribute, reaches 47.7 WGA. LatentRisk-VLM reaches 64.1 WGA without attribute labels, indicating that the useful failure-derived edit is not simply identical to removing the globally most attribute-predictive direction. The direction-alignment file reports normalized Frobenius overlaps of 0.0008 and 0.0120 for the two classes, so the measured case falls into the low-alignment regime: LatentRisk-VLM captures failure factors beyond the named background attribute, or captures them in a class-conditional form not matched by global erasure.

Group-Free Parameter Selection

Table 4 shows that label-free selection, not only direction discovery, determines the final robustness result. Worst-class validation accuracy is the strongest group-free selector in both datasets, recovering most of the Waterbirds oracle but still leaving a 9.4-point CelebA gap to validation-WGA selection. Overall task accuracy preserves average accuracy but selects much weaker WGA operating points. The high-risk, net-correction, and combined proxies are deterministic and group-free, but in these runs they overemphasize examples that do not match the held-out worst groups. Thus, proxy design remains an open part of the method rather than a solved replacement for validation groups.

Retrieval Skew under Demographic Audit

Table 5 shows that LatentRisk-VLM reduces FairFace MaxSkew from 0.355 to 0.323 while Flickr30K R@10 falls by 4.0 points. Projection prompts obtain a similar skew reduction without a utility loss. BEND-VLM obtains the lowest skew using gender-labeled FairFace training references. SANER uses predefined gender terms but no per-example attribute labels; it reduces skew further than LatentRisk-VLM while losing more retrieval utility. This binary-gender, 25-query audit does not establish general demographic debiasing.

Ablation Study

Table 6 shows that semantic residualization is necessary on Waterbirds under the macro-accuracy ablation selector: removing it lowers average accuracy from 76.3 to 66.6 and WGA from 54.3 to 37.4. Other components are not uniformly beneficial. Removing geometric suppression improves WGA from 54.3 to 57.1, and removing reinjection changes WGA by only 0.1 point. Validation macro accuracy selects $\lambda = 0$ for the full rank-4 editor, making it identical to the no-risk-penalty row. The selector sweep in Table 4 shows that this

Table 2: Waterbirds WGA under an attribute-label budget. Values are mean ± standard deviation over three sampled budgets/seeds. LatentRisk-VLM-UL uses no attribute labels, so the result is constant across budgets. CnC-Frozen is rerun with a budget-aware calibration selector on cached CLIP features; it is therefore a budget-sweep adaptation result rather than a copied Table 1 entry.

Method	0	100	500	full
Frozen CLIP	33.6 ± 0.0	33.6 ± 0.0	33.6 ± 0.0	33.6 ± 0.0
LatentRisk-VLM-UL	64.1 ± 0.0	64.1 ± 0.0	64.1 ± 0.0	64.1 ± 0.0
LEACE	N/A	41.0 ± 4.3	33.1 ± 1.7	33.6 ± 0.0
INLP	N/A	44.5 ± 1.0	41.6 ± 1.5	43.1 ± 0.0
DFR-Frozen	N/A	78.9 ± 7.7	88.1 ± 0.8	90.2 ± 0.2
CnC-Frozen	N/A	33.6 ± 0.0	33.6 ± 0.0	33.6 ± 0.0

Table 3: Waterbirds concept-erasure comparison on cached CLIP features. LEACE/INLP use attribute labels to learn a global erasure direction. LatentRisk-VLM uses task labels and failure signals instead. Attribute-probe accuracy is measured after editing; TODO means that the corresponding probe was not defined by the LatentRisk artifact rather than filled with an assumed value.

Method	Direction source	Attr. labels	WGA	Avg.	Attr. probe after
Frozen CLIP	none	0	33.6 ± 0.0	83.7 ± 0.0	94.9 ± 0.0
LEACE	closed-form attribute-predictive linear subspace	1199	33.6 ± 0.0	83.7 ± 0.0	94.9 ± 0.0
INLP	iterative linear probes for attribute predictability	1199	43.1 ± 0.0	85.5 ± 0.0	70.6 ± 0.0
LatentVLM	failure-derived latent risk direction	0	64.1 ± 0.0	79.9 ± 0.0	TODO
Random direction	random orthonormal direction sanity check	0	33.4 ± 0.9	83.9 ± 0.2	94.9 ± 0.0
Oracle mean-difference	oracle attribute mean-difference direction	1199	47.7 ± 0.0	83.8 ± 0.0	53.8 ± 0.0

is a selection failure rather than conclusive evidence against rank four: worst-class selection raises the same rank-4 editor to 63.1 WGA.

Risk Separability Is Insufficient

On the calibration split, the Waterbirds mean-contrast directions separate risk labels with AUCs of 0.977 and 0.932 for landbirds and waterbirds. The CelebA rank-1 probes reach 0.989 and 0.981 for non-blond and blond classes. Yet the edits have opposite WGA effects. These are in-sample diagnostics, not generalization estimates, but they establish an important negative result: making failure linearly separable does not show that the direction corresponds to a hidden group or that removing it will improve group robustness.

Post-hoc Alignment Diagnostics

Table 7 directly tests the condition that failure-derived directions should overlap the hidden group factor. For each class c, we compute a post-hoc spurious direction v_c as the normalized difference between the two group-conditioned mean CLIP image embeddings on the calibration split, and compare it with the learned rank-1 risk direction u_c using $|u_c^\top v_c|$. Group labels are used only for this diagnostic, not for risk estimation or editing. Waterbirds has substantially higher alignment than CelebA (0.420 versus 0.080), and its detected high-risk set is enriched for minority-group samples: 69.2% of top-risk calibration examples are minority examples, compared with a 50.0% minority base rate. CelebA shows the opposite pattern. Its top-risk set contains 40.0% minority examples, below the 42.6% base rate, so the risk score does not identify the Male-defined worst-group structure.

The label-free diagnostics refine this interpretation. Error residuals are slightly more concentrated on Waterbirds than on CelebA (0.086 versus 0.072), matching the direction of the WGA gains. Stability alone is not a sufficient reliability criterion: CelebA directions are highly reproducible across random half-splits (0.893), but they are reproducibly aligned with non-gender error structure. Thus, the CelebA failure is not caused by an unstable estimator. It is caused by a stable but misaligned risk direction: suppressing it raises average accuracy while damaging the group that defines WGA. The diagnostic supports the paper's central claim that latent failures provide useful weak supervision only when the failure direction is also a group-relevant direction.

Backbone Transfer

We next ask whether the Waterbirds–CelebA contrast depends on the CLIP ViT-L/14 backbone. We repeat only the

Table 4: Rank-4 LatentRisk-VLM selection over the same candidate pool. All non-oracle selectors use validation task labels and frozen-baseline scores, but no group attributes. The oracle selector uses validation WGA and is included only as an upper bound.

	Selected candidate		Waterbirds			CelebA		
Selector	Waterbirds	CelebA	WGA	Avg.	Gap	WGA	Avg.	Gap
Task accuracy	$\lambda = 0.085, b = 0.004$	$\lambda = 0.050, b = -0.019$	46.0	83.9	38.0	52.2	91.6	39.4
Worst-class accuracy	$\lambda = 0.070, b = 0.018$	$\lambda = 0.085, b = -0.007$	63.1	79.5	16.4	72.8	87.9	15.1
High-risk accuracy	$\lambda = 0.110, b = 0.030$	$\lambda = 0.110, b = -0.030$	41.8	70.5	28.7	23.9	90.3	66.4
Net correction	$\lambda = 0.070, b = 0.025$	$\lambda = 0.080, b = -0.030$	51.4	75.1	23.7	22.2	90.3	68.1
Combined label-free	$\lambda = 0.100, b = 0.028$	$\lambda = 0.055, b = -0.023$	45.5	72.3	26.8	41.7	91.4	49.7
Validation-WGA oracle	$\lambda = 0.085, b = 0.017$	$\lambda = 0.300, b = -0.001$	64.1	79.9	15.8	82.2	85.7	3.5

Table 5: Measured retrieval with frozen CLIP ViT-L/14. MS@100 is mean gender MaxSkew@100 over 25 FairFace stereotype queries (lower is better). Flickr30K reports text-to-image Recall@10 on the Karpathy 1K test split (higher is better).

Method	MS@100 ↓	R@10 ↑
Frozen CLIP	0.355	92.2
Projection prompts	0.326	**92.3**
BEND-VLM	**0.141**	91.6
SANER	0.202	85.2
LatentRisk-VLM	0.323	88.2

Table 6: Waterbirds ablation with frozen CLIP ViT-L/14. Columns y and a indicate task- and attribute-label use. The last row is not a primary result.

Variant	y	a	Avg.	WGA
Frozen CLIP	N	N	83.7	33.6
SPD (oracle attr.)	N	Y	77.2	22.7
Single mean-contrast direction	Y	N	78.4	60.5
No semantic residualization	Y	N	66.6	37.4
No geometric suppression	Y	N	77.0	57.1
No risk-score penalty	Y	N	76.3	54.3
No neutral reinjection	Y	N	76.3	54.4
Full rank-4 editor (macro sel.)	Y	N	76.3	54.3
Rank 4 + attr.-oracle selection	Y	Y	**79.9**	**64.1**

low-cost pieces of the experiment with frozen SigLIP Base Patch16/224 (Zhai et al. 2023): frozen zero-shot classification, rank-1 editing, rank-4 editing with the group-free worst-class selector, and a rank-4 validation-WGA oracle used only as a diagnostic upper bound. The encoders remain frozen, all image and text embeddings are normalized before editing, and WGA gains are computed relative to the frozen model with the same backbone.

Table 8 gives a deliberately mixed transfer result. The qualitative claim that task-accuracy selection is unsafe transfers: on both Waterbirds and CelebA, SigLIP rank-1 editing increases or preserves average accuracy while driving WGA below the frozen SigLIP baseline. On Waterbirds, the rank-1 risk direction is highly aligned with the place direction and the high-risk set is strongly minority-enriched, yet the selected candidate catastrophically shifts predictions toward the wrong group. Rank-4 worst-class selection partially corrects this, improving WGA from 8.1 to 22.0, and the validation-WGA oracle raises it only to 25.9. Thus, with SigLIP the diagnostic still identifies group-relevant failure structure, but candidate selection and semantic preservation become the limiting factors.

CelebA shows a different transfer pattern. The SigLIP frozen classifier has high average accuracy but extremely low Blond-Hair WGA, making the baseline heavily group-skewed. Rank-1 task-accuracy selection again fails, reaching 0.0 WGA. Rank-4 worst-class selection raises WGA to 55.0, but at a large average-accuracy cost. The alignment diagnostic remains low (0.057), matching the CLIP conclusion that the failure-derived risk direction is not a reliable Male-direction surrogate. The positive rank-4 WGA gain should therefore be read as a correction of a severe frozen-SigLIP operating point, not as evidence that risk directions recover demographic attributes.

Table 9 controls the rank-1 estimator choice. In these SigLIP runs, mean contrast and a one-step logistic probe select nearly identical task-accuracy operating points and produce the same WGA failures. The CLIP Waterbirds–CelebA contrast is therefore not explained solely by using mean contrast on one dataset and a probe on the other; the downstream selector and backbone-specific zero-shot geometry matter.

Predicting Editing Outcomes on a New Shift

To test the diagnostic protocol on a third setting, we construct a MetaShift-style cat-versus-dog shift using GQA scene graphs (Hudson and Manning 2019) linked to Visual Genome images (Krishna et al. 2017), following the spirit of MetaShift's contextual-shift evaluation (Liang and Zou 2022). We use indoor/outdoor context as the group attribute. The metadata contains 1,944 images: 1,500 train, 300 validation/calibration, and 144 held-out test examples. The training split is spuriously correlated, with cat–indoor and dog–outdoor majority cells of 600 images and minority

Table 7: Diagnostics explaining why rank-1 latent-risk editing helps Waterbirds but hurts CelebA. Alignment is the average over classes of the absolute cosine similarity between the learned risk direction and a post-hoc spurious direction computed from group-labeled calibration embeddings. Risk-minority share uses the top 20% calibration samples by class-conditional risk-direction score. Stability and concentration are label-free reliability diagnostics; WGA gain is measured on the test split against frozen CLIP.

Dataset	Align.	Risk-min.	Overall min.	Enrich.	Recall min.	Stability	Concentr.	WGA gain
Waterbirds	0.420	69.2	50.0	1.38	27.7	0.751 ± 0.016	0.086	+28.2
CelebA	0.080	40.0	42.6	0.94	18.8	0.893 ± 0.007	0.072	-17.1

Table 8: Backbone transfer results. WGA, Avg., Gap, and WGA gain are percentage points. Alignment and enrichment are calibration diagnostics: alignment is the post-hoc absolute cosine similarity between the risk direction and the group-defined spurious direction, while enrichment is the top-risk minority share divided by the calibration minority base rate. Dashes denote quantities that are not defined for the frozen row. Oracle rows use validation group labels only for model selection and are diagnostic upper bounds, not group-free results.

Backbone	Dataset	Method	Selector	WGA	Avg.	Gap	Align.	Enrich.	WGA gain
CLIP	Waterbirds	Frozen	none	33.6	83.7	50.1	–	–	0.0
CLIP	Waterbirds	rank 1	reported	60.5	78.4	17.9	0.420	1.38	+26.9
SigLIP	Waterbirds	Frozen	none	8.1	27.3	19.2	–	–	0.0
SigLIP	Waterbirds	rank 1	task accuracy	0.2	75.9	75.7	0.499	1.74	-7.9
SigLIP	Waterbirds	rank 4	worst-class	22.0	44.9	22.9	0.366	0.35	+14.0
SigLIP	Waterbirds	rank 4 oracle	validation WGA	25.9	42.7	16.9	0.366	0.35	+17.8
CLIP	CelebA	Frozen	none	71.5	78.2	6.7	–	–	0.0
CLIP	CelebA	rank 1	reported	54.4	91.7	37.3	0.080	0.94	-17.1
SigLIP	CelebA	Frozen	none	6.0	85.1	79.1	–	–	0.0
SigLIP	CelebA	rank 1	task accuracy	0.0	86.7	86.7	0.057	1.15	-6.0
SigLIP	CelebA	rank 4	worst-class	55.0	56.2	1.2	0.057	1.15	+49.0
SigLIP	CelebA	rank 4 oracle	validation WGA	55.0	56.2	1.2	0.057	1.15	+49.0

Table 9: Estimator control on SigLIP rank-1 runs. Both estimators use task labels only and task-accuracy selection. WGA and Avg. are percentages.

Dataset	Estimator	WGA	Avg.	WGA gain
Waterbirds	mean contrast	0.2	75.9	-7.9
Waterbirds	logistic probe	0.2	75.9	-7.9
CelebA	mean contrast	0.0	86.7	-6.0
CelebA	logistic probe	0.0	86.7	-6.0

cells of 150 images; validation has 75 examples per cell and test has 36 examples per cell. The subset is deterministic and uses natural prompts “a photo of a cat” and “a photo of a dog.” Because the construction uses a keyword heuristic over scene-graph object/context names rather than the official MetaShift repository pipeline, we treat it as a MetaShift-style diagnostic shift rather than an exact benchmark reproduction.

Before held-out test editing, we run calibration-only risk discovery and write a preregistered prediction file. The rule is intentionally transparent: a likely-success prediction requires high alignment, minority enrichment above one, minority recall above the configured floor, sufficient split-half stability, and nontrivial risk concentration. Low alignment or no minority enrichment predicts likely failure; conflicting evidence predicts uncertainty. The thresholds are fixed in the experiment configuration and the generated prediction records `test_performance_used=false`.

Tables 10 and 11 close the diagnostic loop. The calibration diagnostics are mixed: alignment and enrichment pass the success-side thresholds, but split-half stability is low. The rule therefore predicts *uncertain*, not likely success. On the held-out test split, rank-4 editing raises WGA from 19.4 to 25.0 while reducing average accuracy from 61.8 to 58.3.

Table 10: Diagnostic prediction and held-out outcome on the MetaShift-style cat/dog shift with frozen SigLIP. The prediction is generated before final test editing. WGA gain is relative to frozen SigLIP on the same test split.

Dataset	Align.	Enrich.	Recall min.	Stability	Concentr.	Risk AUC	Observed gain	Prediction	Correct?
MetaShift-style cat/dog	0.226	1.33	0.267	0.267	0.093	0.635	+5.6	uncertain	n/a

Table 11: Held-out MetaShift-style classification with frozen SigLIP. Values are percentages.

Method	WGA	Avg.	Gap	WGA gain
Frozen SigLIP	19.4	61.8	42.4	0.0
LatentRisk-VLM rank 4	25.0	58.3	33.3	+5.6

This positive but small WGA gain is compatible with the uncertainty prediction: the direction contains some group-relevant signal, but its instability and modest concentration warn against expecting a strong or reliable improvement. Per-group accuracy after editing is 72.2 for cat–indoor, 25.0 for cat–outdoor, 58.3 for dog–indoor, and 77.8 for dog–outdoor, so the remaining worst group is the cat–outdoor minority cell.

Computational Cost of Frozen-Feature Editing

The extension also illustrates the cost profile of the method. Changing the backbone requires one frozen feature-extraction pass; all subsequent risk discovery, candidate scoring, diagnostics, and table generation operate on cached embeddings. The SigLIP embedding caches are 39 MB for Waterbirds, 218 MB for CelebA, and 2.5 MB for the MetaShift-style subset. Once cached, feature loading for method runs takes seconds: 2.9–3.0 seconds for Waterbirds, 16.3–16.4 seconds for CelebA, and 0.34 seconds for MetaShift. The editing/search/evaluation stage takes 9.4–9.5 seconds on Waterbirds, 95–101 seconds on CelebA, and 40.7 seconds on MetaShift under the reported grids. These timings support the intended low-cost interpretation for the proposed editor, but they should not be used to claim that CnC-Frozen or DFR-Frozen retrain a full vision backbone in our codebase. The revised efficiency analysis reports end-to-end feature extraction separately from post-feature adaptation and reads repeated-run wall-clock and memory measurements from run artifacts before any paper table is filled.

Table 12 makes the revised comparison explicit. DFR-Frozen is both more accurate and cheaper than LatentRisk-VLM in this Waterbirds feature-space refit, but it uses attribute labels and trains a classifier head. The budget-aware CnC-Frozen adapter optimizes many more feature-space parameters and takes 236.7 seconds on average in the CPU PyTorch rerun, yet remains at 33.6 WGA in this particular calibration-budget protocol. LEACE and INLP solve or fit only small feature-space erasers; in our runs their robustness gains are much smaller than DFR-Frozen and smaller than LatentRisk-VLM. The correct efficiency claim is therefore not that LatentRisk-VLM is universally cheaper or stronger than supervised frozen-feature baselines. It is that LatentRisk-VLM runs after feature extraction, updates no backbone or classifier head, and remains available when the attribute-label budget is zero.

Discussion and Limitations

Failure–group alignment is the key assumption. The method never identifies a group as such. It identifies representation variation associated with low margins within a class. Worst-group performance improves only when that variation overlaps the factor defining the worst group and can be removed without erasing class evidence. The alignment diagnostic in Table 7 makes this assumption measurable: Waterbirds has higher risk–spurious alignment and its top-risk set is minority-enriched, whereas CelebA has low gender-direction alignment and no minority enrichment. The SigLIP and MetaShift-style extensions reinforce the same caveat. High or moderate alignment can still be insufficient if candidate selection destroys class evidence or if split-half stability is low. CelebA therefore fails not because the risk direction is noisy, but because it is stable and points at other error structure. Accordingly, LatentRisk-VLM should be described as latent-risk editing, not unsupervised demographic debiasing.

Discovery and selection are distinct. Tables 1 and 4 separate group-label use during adaptation from group-label use during model selection. The CelebA oracle result is informative precisely because it is not a primary result: group labels are absent from subspace discovery in both the 72.8- and 82.2-WGA rows, yet validation-WGA selection still produces a 9.4-point WGA gain over the best group-free selector. Thus, eliminating group labels from representation discovery does not by itself yield a fully group-free pipeline. A deployment-relevant selector must also choose rank and scoring parameters without validation groups. Future work should target group-label-free robust selection, for example through stability across environments or tail-risk criteria, rather than assuming average accuracy will choose a group-robust edit.

Selection supervision and adaptation supervision should not be conflated. Methods such as AFR, CnC-Frozen, CFR, and GIC avoid true group labels in their fitting stage in the reported protocol but use validation groups to choose checkpoints or hyperparameters. SPD and BEND-VLM use attribute information earlier, during direction or reference construction. These are materially different supervision regimes, even though both can improve WGA. CnC-Frozen outperforms our CelebA oracle under group-aware validation, showing that reliable validation groups remain powerful when available. Latent-risk editing addresses an unnamed-nuisance setting and should not be presented as a replacement for methods that can legitimately use group supervision.

Table 12: Post-feature adaptation cost on Waterbirds with cached frozen CLIP features. Feature extraction is excluded from adaptation time. Values are mean $\pm$ standard deviation over three seeds when available. CnC-Frozen is measured using the budget-aware cached-feature adapter from the attribute-label sweep, not copied from the single Table 1 configuration.

Method	Attr. labels	Optimized object	Params	Adapt. time (s)	WGA	Avg.
Frozen zero-shot CLIP	No	none	0	0.000 ± 0.000	33.6 ± 0.0	83.7 ± 0.0
LatentVLM	No	class-conditional low-rank risk directions and scoring grid	4608	11.005 ± 0.546	64.1 ± 0.0	79.9 ± 0.0
CnC-Frozen	Yes	feature-space linear encoder adapter and classifier with contrastive/classification loss	591362	236.699 ± 145.405	33.6 ± 0.0	83.8 ± 0.0
DFR-Frozen	Yes	feature-space classifier refit	769	0.121 ± 0.024	90.2 ± 0.2	91.4 ± 0.9
LEACE	Yes	closed-form attribute-predictive linear subspace	768	0.271 ± 0.114	33.6 ± 0.0	83.7 ± 0.0
INLP	Yes	iterative linear probes for attribute predictability	768	0.100 ± 0.007	43.1 ± 0.0	85.5 ± 0.0

Complementary operating regime. LatentRisk-VLM targets a complementary operating regime: frozen vision-language models with task labels but without sensitive-attribute annotations. Our goal is not to replace fully supervised group-robust training when dense group annotations and retraining resources are available. Instead, we study whether model failures themselves can provide a low-cost signal for post-hoc risk suppression in frozen VLM representations. The Waterbirds budget experiment confirms that explicit attribute labels can be much stronger: DFR-Frozen reaches 90.2 WGA with the full validation attribute set, compared with 64.1 for LatentRisk-VLM-UL. This result should be reported directly. The relevant claim for LatentRisk-VLM is not dominance over attribute-supervised methods, but viability at the zero-attribute-label point and a diagnostic account of when failure supervision is informative.

Experimental limitations. The added SigLIP runs broaden the backbone evidence but do not establish architecture-level robustness: they cover one additional backbone, one deterministic seed for main results, and only the minimal frozen/rank-1/rank-4 matrix. The MetaShift-style cat/dog setting is constructed from GQA scene graphs and Visual Genome images with a deterministic keyword heuristic; it should not be treated as a full official MetaShift benchmark reproduction. The classification datasets remain binary, and the retrieval audit reduces gender to the binary labels supplied by FairFace and uses only 25 stereotype queries. The measured rank-1 estimator differs across the original CLIP classification datasets (mean contrast on Waterbirds and a logistic probe on CelebA), although the SigLIP estimator-control runs suggest that this choice is not the sole driver of the observed rank-1 failures. The alignment diagnostic itself uses group labels post hoc and is explanatory rather than deployable; the concentration and stability diagnostics are group-free but are still measured on a small number of settings. Several baselines are adaptations to frozen CLIP features rather than exact reproductions in their original architectures and data regimes. The same validation split is used for risk discovery and parameter selection, which can favor calibration-specific structure. Finally, a predictive direction is correlational: these experiments do not establish that it causally generates errors or that editing it is safe under unmeasured shifts.

Ethical use. An attribute-free method can still worsen a protected group, as the CelebA result demonstrates. Aggregate accuracy must not be used to certify fairness, and post-hoc subgroup audits remain necessary whenever they are appropriate and lawful. The method should not be deployed to remove demographic information indiscriminately or to infer sensitive attributes. Its appropriate role is as an auditable representation intervention whose group consequences are measured explicitly.

Conclusion

We studied whether frozen-VLM failures can guide representation editing without group annotations. LatentRisk-VLM uses task-label margins, class residualization, risk-direction estimation, and candidate-wise inference without updating either encoder. It discovers error-related latent risk directions, not sensitive attributes or demographic groups. Rank-1 editing substantially improves CLIP Waterbirds WGA, while rank-4 editing with worst-class validation selection improves both CLIP Waterbirds and CLIP CelebA over task-accuracy selection. The SigLIP extension and the MetaShift-style cat/dog shift add a more cautious picture: changing the frozen backbone or shift can make task-accuracy selection fail badly, and rank-4 selection may recover WGA only with substantial average-accuracy tradeoffs or modest gains. Post-hoc diagnostics explain why rank-1 CelebA fails: the learned risk direction is stable but poorly aligned with the Male-defined spurious direction, and top-risk examples are not minority-enriched. The revised protocol positions CnC-Frozen, DFR-Frozen, INLP, and LEACE as supervised or attribute-budgeted frozen-feature adaptations rather than as full-backbone training costs. Methods with explicit attribute labels or group-aware validation remain stronger in settings where that supervision is available. Failure-derived directions are useful weak supervision when failure and hidden groups align, not a general substitute for group information.

References

Agarwal, S.; Krueger, G.; Clark, J.; Radford, A.; Kim, J. W.; and Brundage, M. 2021. Evaluating CLIP: Towards Characterization of Broader Capabilities and Downstream Implications. *arXiv preprint arXiv:2108.02818*.

Belrose, N.; Schneider-Joseph, D.; Ravfogel, S.; Cotterell, R.; Raff, E.; and Biderman, S. 2023. LEACE: Perfect Linear Concept Erasure in Closed Form. In *Advances in Neural Information Processing Systems*, volume 36.

Berg, H.; Hall, S.; Bhalgat, Y.; Kirk, H.; Shtedritski, A.; and Bain, M. 2022. A Prompt Array Keeps the Bias Away: Debiasing Vision-Language Models with Adversarial Learning. In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing*, 806–822.

Chuang, C.-Y.; Jampani, V.; Li, Y.; Torralba, A.; and Jegelka, S. 2023. Debiasing Vision-Language Models via Biased Prompts. *arXiv preprint arXiv:2302.00070*.

Creager, E.; Jacobsen, J.-H.; and Zemel, R. 2021. Environment Inference for Invariant Learning. In *International Conference on Machine Learning*, 2189–2200.

Gerych, W.; Zhang, H.; Hamidieh, K.; Pan, E.; Sharma, M.; Hartvigsen, T.; and Ghassemi, M. 2024. BEND-VLM: Test-Time Debiasing of Vision-Language Embeddings. In *Advances in Neural Information Processing Systems*.

Hall, M.; Gustafson, L.; Adcock, A.; Misra, I.; and Ross, C. 2023. Vision-Language Models Performing Zero-Shot Tasks Exhibit Disparities Between Gender Groups. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2778–2785.

Hamidieh, K.; Zhang, H.; Gerych, W.; Hartvigsen, T.; and Ghassemi, M. 2024. Identifying Implicit Social Biases in Vision-Language Models. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 547–561.

Hirota, Y.; Chen, M.-H.; Wang, C.-Y.; Nakashima, Y.; Wang, Y.-C. F.; and Hachiuma, R. 2025. SANER: Annotation-Free Societal Attribute Neutralizer for Debiasing CLIP. In *International Conference on Learning Representations*.

Hudson, D. A.; and Manning, C. D. 2019. GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 6700–6709.

Jung, H.; Jang, T.; and Wang, X. 2024. A Unified Debiasing Approach for Vision-Language Models across Modalities and Tasks. In *Advances in Neural Information Processing Systems*.

Karkkainen, K.; and Joo, J. 2021. FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age for Bias Measurement and Mitigation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 1548–1558.

Karpathy, A.; and Fei-Fei, L. 2015. Deep Visual-Semantic Alignments for Generating Image Descriptions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 3128–3137.

Kirichenko, P.; Izmailov, P.; and Wilson, A. G. 2023. Last Layer Re-Training Is Sufficient for Robustness to Spurious Correlations. In *International Conference on Learning Representations*.

Krishna, R.; Zhu, Y.; Groth, O.; Johnson, J.; Hata, K.; Kravitz, J.; Chen, S.; Kalantidis, Y.; Li, L.-J.; Shamma, D. A.; Bernstein, M. S.; and Fei-Fei, L. 2017. Visual Genome: Connecting Language and Vision Using Crowdsourced Dense Image Annotations. *International Journal of Computer Vision*, 123(1): 32–73.

Liang, W.; and Zou, J. 2022. MetaShift: A Dataset of Datasets for Evaluating Contextual Distribution Shifts and Training Conflicts. In *International Conference on Learning Representations*.

Liu, E. Z.; Haghgoo, B.; Chen, A. S.; Raghunathan, A.; Koh, P. W.; Sagawa, S.; Liang, P.; and Finn, C. 2021. Just Train Twice: Improving Group Robustness without Training Group Information. In *International Conference on Machine Learning*, 6781–6792.

Liu, Z.; Luo, P.; Wang, X.; and Tang, X. 2015. Deep Learning Face Attributes in the Wild. In *Proceedings of the IEEE International Conference on Computer Vision*, 3730–3738.

Qiu, S.; Potapczynski, A.; Izmailov, P.; and Wilson, A. G. 2023. Simple and Fast Group Robustness by Automatic Feature Reweighting. *arXiv preprint arXiv:2306.11074*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. 2021. Learning Transferable Visual Models from Natural Language Supervision. In *International Conference on Machine Learning*, 8748–8763.

Ravfogel, S.; Elazar, Y.; Gonen, H.; Twiton, M.; and Goldberg, Y. 2020. Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 7237–7256.

Sagawa, S.; Koh, P. W.; Hashimoto, T. B.; and Liang, P. 2020. Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. In *International Conference on Learning Representations*.

You, C.; Min, Y.; Dai, W.; Sekhon, J. S.; Staib, L.; and Duncan, J. S. 2024. Calibrating Multi-modal Representations: A Pursuit of Group Robustness without Annotations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Young, P.; Lai, A.; Hodosh, M.; and Hockenmaier, J. 2014. From Image Descriptions to Visual Denotations: New Similarity Metrics for Semantic Inference over Event Descriptions. *Transactions of the Association for Computational Linguistics*, 2: 67–78.

Zhai, X.; Mustafa, B.; Kolesnikov, A.; and Beyer, L. 2023. Sigmoid Loss for Language Image Pre-Training. *arXiv preprint arXiv:2303.15343*.

Zhang, M.; Sohoni, N. S.; Zhang, H. R.; Finn, C.; and Ré, C. 2022. Correct-N-Contrast: A Contrastive Approach for Improving Robustness to Spurious Correlations. In *International Conference on Machine Learning*.

Zhao, D.; Li, W.; Shen, Z.; Qiu, Y.; Xu, B.; Chen, H.; and Chen, Y. 2026. Bias Is a Subspace, Not a Coordinate: A Geometric Rethinking of Post-hoc Debiasing in Vision-Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Zhou, K.; Lai, E.; and Jiang, J. 2022. VLStereoSet: A Study of Stereotypical Bias in Pre-trained Vision-Language Models. In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics* *and the 12th International Joint Conference on Natural Language Processing*, 527–538.